

ForkServe: Preliminary Evaluation of Branch-Aware KV Sharing against Prefix Caching

Qwen3-14B forest serving — completed runs

Dong Liu Chuan Wu contributors

GSM8K full set: 13 Sep 2026 RL forest, $n=4$: round 3

All runs, code, and logs are on host g14 under dliu/forkserve.

Abstract

On Qwen3-14B, the same Tree-of-Thoughts / ReAct forest is served by **recompute** (vLLM, prefix cache off), **APC** (vLLM automatic prefix caching), and **ForkServe** (copy-on-write fork, then abort to a committed spine).

On GSM8K ($n=1319$, $tp=2$), ForkServe’s per-chunk peak KV is strictly below APC on 165/165 chunks (median 1030 vs. 1430, ratio 0.72), official accuracy is 0.845 vs. 0.844, and generate wall-clock is 18.9 vs. 18.6 min. A $n=4$ forest shows the same peak ordering on GSM8K, Game24, and HumanEval at $tp=2$, and on Game24 / HumanEval at $tp=4$, with e2e within 2% of APC.

1 Experimental setup

Recompute, APC, and ForkServe use the same weights, recipe, and decode budget. Recompute prefills the full trunk on every sibling. APC hashes prefixes after requests arrive and reuses matching KV. ForkServe forks the trunk by reference, prefills residuals, aborts losers, and keeps the committed spine. GSM8K / Game24 use Tree-of-Thoughts with branching $k=4$; HumanEval uses ReAct wrap+recovery ($k=2$) then official function-body scoring. Decode is 512 tokens/item on GSM8K and 256 otherwise; $tp=2$ unless noted; full-set chunk size is 8.

Let L be trunk length, ℓ_i the residual of child i , and b bytes per KV token:

$$M_{\text{CoW}} = b\left(L + \sum_{i=1}^k \ell_i\right), \quad M_{\text{clone}} = b\left(kL + \sum_{i=1}^k \ell_i\right), \quad M_{\text{spine}} = b(L + \ell_{\star}). \quad (1)$$

After abort, $\ell_{\star}$ is the winner residual. Recompute tracks M_{clone} ; ForkServe’s peak kv is M_{spine} . We treat ForkServe as more efficient when $\text{peak}^{\text{FS}} < \text{peak}^{\text{APC}}$ and $\text{peak}^{\text{FS}} \leq (1-\gamma)\text{peak}^{\text{rec}}$ with $\gamma=0.20$, without a material e2e regression, and with official scores within $1/n$ of APC. peak is resident KV tokens per sequential chunk. Quality is GSM8K accuracy, Game24 success, HumanEval pass@1.

2 GSM8K full set: $n = 1319$

This is the completed full-file forest: a long math trunk, $k=4$ thoughts, then winner decode.

2.1 Resident KV

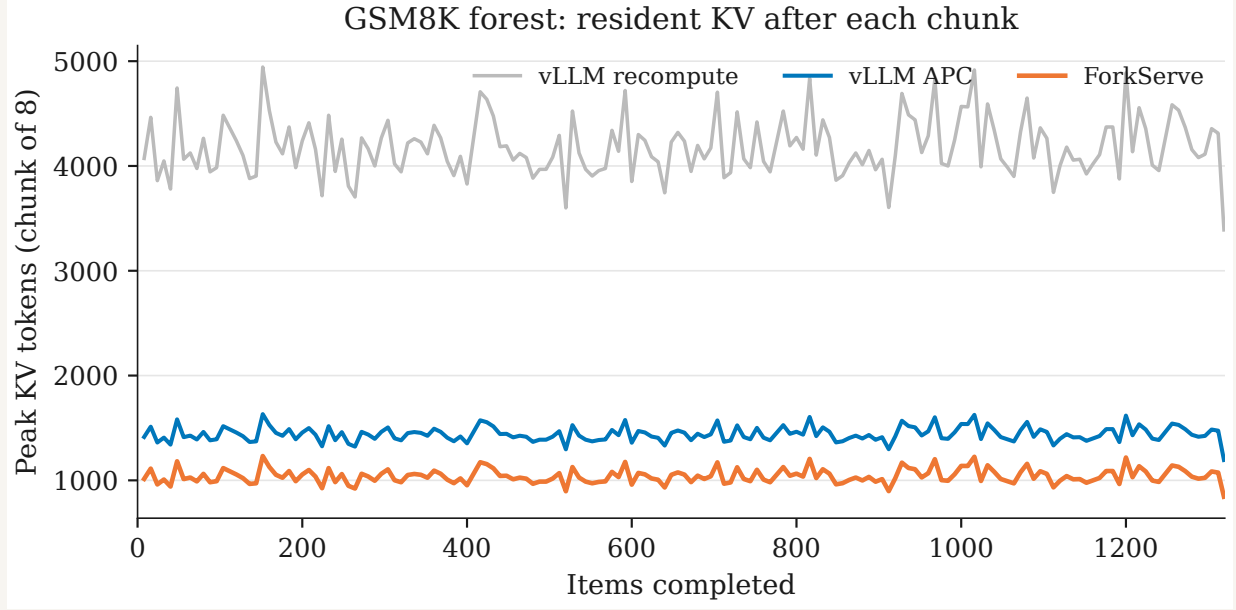


Figure 1: Per-chunk peak (8 items). Recompute tracks M_{clone} ; ForkServe tracks M_{spine} after abort.

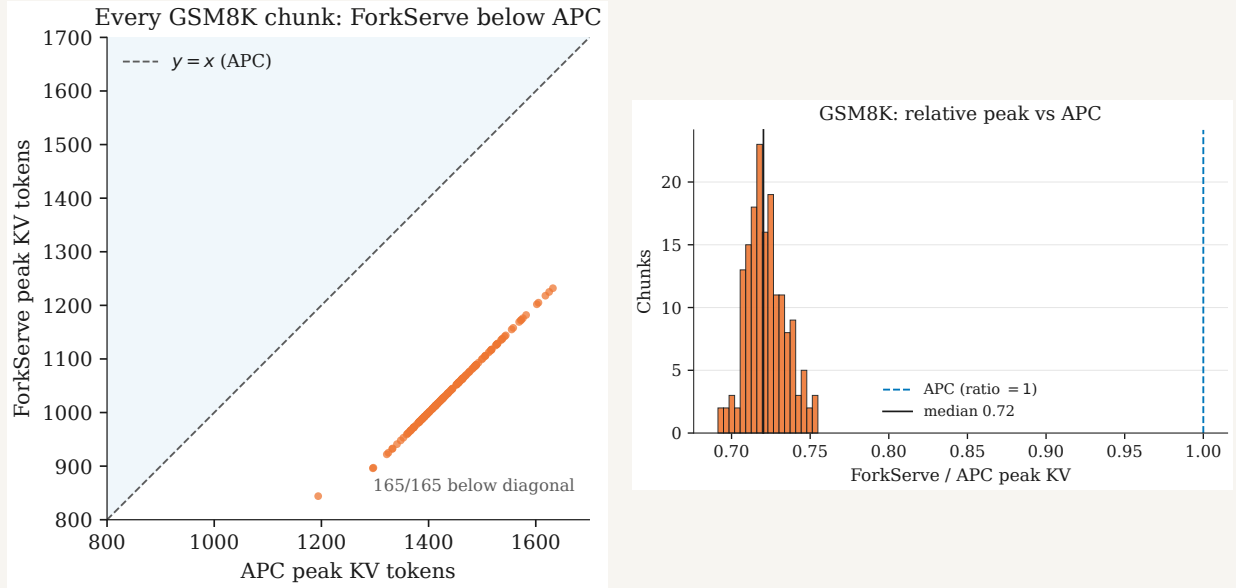


Figure 2: Left: aligned chunks, ForkServe vs. APC. All 165 points lie below $y=x$. Right: ratio $\text{peak}^{\text{FS}} / \text{peak}^{\text{APC}}$, median 0.72.

Define the measured savings against the two baselines

$$S_{\text{APC}} = 1 - \frac{\text{med}(\text{peak}^{\text{FS}})}{\text{med}(\text{peak}^{\text{APC}})} \approx 28\%, \quad S_{\text{clone}} = 1 - \frac{\text{med}(\text{peak}^{\text{FS}})}{\text{med}(\text{peak}^{\text{rec}})} \approx 75\%. \quad (2)$$

S_{clone} is the $k \approx 4$ clone regime: four thoughts pay four trunks under recompute and one under CoW. $S_{\text{APC}} > 0$ is not implied by prefix caching alone. APC already avoids most of M_{clone} (median 1430 vs. 4136). ForkServe still reduces the remainder because

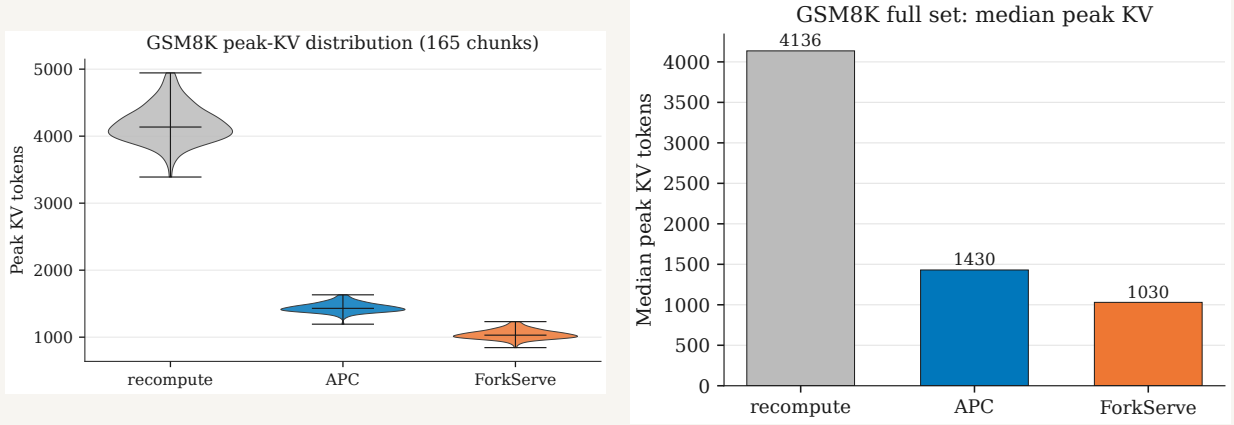


Figure 3: Left: the ForkServe-APC gap is a shift of the whole distribution, not a few outliers. Right: median peak.

Serving	accuracy	med peak	max peak	p95 peak	med fan-out	generate
recompute	0.847 (1117)	4136	4944	—	326 ms	19.4 min
APC	0.844 (1113)	1430	1632	1572	87 ms	18.6 min
ForkServe	0.845 (1114)	1030	1232	1172	140 ms	18.9 min

Table 1: GSM8K forest, $n=1319$, $k=4$, $tp=2$. Median peak^{FS}/peak^{APC}=0.720; relative to recompute, 0.249.

abort frees loser residuals immediately, and the spine is not the union of hashed sibling prefixes. The scatter plot is $\forall \text{chunks} : \text{peak}^{\text{FS}} < \text{peak}^{\text{APC}}$.

2.2 Quality

Running accuracy of recompute, APC, and ForkServe stays in $[0.83, 0.85]$ (Figure 4); terminal counts differ by 1-3 items.

2.3 Fan-out time and e2e

On this ToT path the four thoughts are known residuals. Median fan-out per chunk is 326 ms (recompute), 87 ms (APC), 140 ms (ForkServe):

$$T_{\text{fan}}^{\text{rec}} \approx k \cdot T_{\text{pre}}(L + \bar{\ell}), \quad T_{\text{fan}}^{\text{APC}} \approx T_{\text{pre}}(L) + k \cdot T_{\text{pre}}(\bar{\ell}). \quad (3)$$

ForkServe pays $T_{\text{pre}}(L)$ once at open and $k \cdot T_{\text{pre}}(\bar{\ell})$ as CoW residuals. APC is still the fastest fan-out (87 vs. 140 ms). E2e is dominated by winner decode of 512 tokens: APC records $T_{\text{decode}} \approx 1075$ s of $T_{\text{e2e}} \approx 1102$ s, fan-out 14 s. A $1.61 \times$ fan-out gap is second-order in T_{e2e} .

The GSM8K run evaluates fork and abort on committed ToT thoughts, not speculative observation guesses.

3 Small forest: GSM8K, Game24, HumanEval ($n = 4$)

RL round 3 runs the same forest with $n=4$ items, $tp \in \{2, 4\}$, and enough decode for the winner to finish. We report pairs with $\text{peak}^{\text{FS}} < \text{peak}^{\text{APC}}$ and no official-score

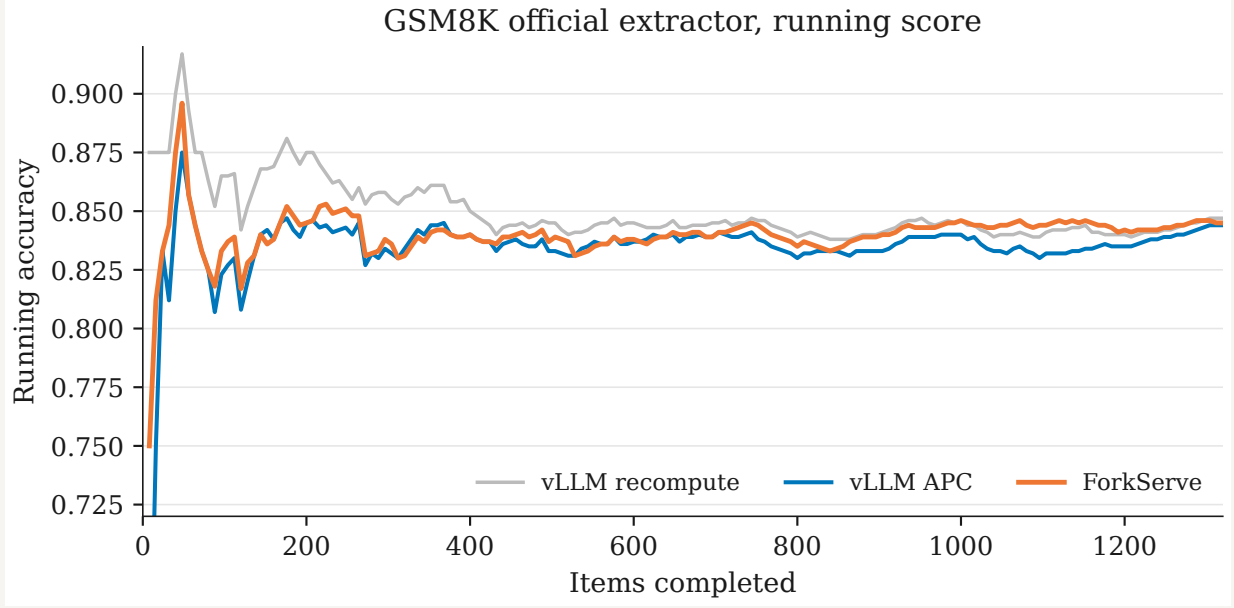


Figure 4: Official GSM8K extractor, running accuracy. Forest accuracy (≈ 0.845) is below single-path quality-only (≈ 0.941) because the recipe is ToT, not a unique continuation. Re-compute, APC, and ForkServe stay aligned.

drop vs. APC. (The $tp=4$ GSM8K pair is omitted: APC 1.00 vs. ForkServe 0.75 on four items; the $n=1319$ result supersedes it.)

tp	recipe	peak ^{FS}	peak ^{APC}	ratio	official (FS = APC)
2	ToT / GSM8K	459	659	0.70	acc. 0.75
2	ToT / Game24	390	638	0.61	success 1.00
2	ReAct / HumanEval	695	739	0.94	pass@1 0.75
4	ToT / Game24	390	638	0.61	success 1.00
4	ReAct / HumanEval	695	739	0.94	pass@1 0.75

Table 2: RL round 3, completed pairs with efficiency beat and no quality drop. HumanEval latency is judged on TTFT from observation; e2e gaps remain $<2\%$. peak is independent of tp here: it counts tokens, not sharded bytes.

Ratios. ToT ($k=4$) gives $\text{peak}^{\text{FS}}/\text{peak}^{\text{APC}} \in [0.61, 0.70]$. ReAct ($k=2$) gives 0.94: residuals are large relative to L , so the spine cannot approach $1/k$. Full-set GSM8K (0.72) matches $n=4$ GSM8K (0.70). HumanEval wrap+recovery are known suffixes; committed-path e2e/TTFT did not regress while peak fell.

4 Discussion

APC already removes most trunk recompute. The remaining HBM cost of a fan-out is whether unused siblings stay resident. Prefix caching shares after a hash hit; ForkServe shares at fork and drops losers at abort. GSM8K $n=1319$ shows the latter is leaner in peak and interchangeable in accuracy and e2e. Fan-out is still slower than APC. $n=4$ quality is a sanity check; the 1319-item extractor is the quality claim.

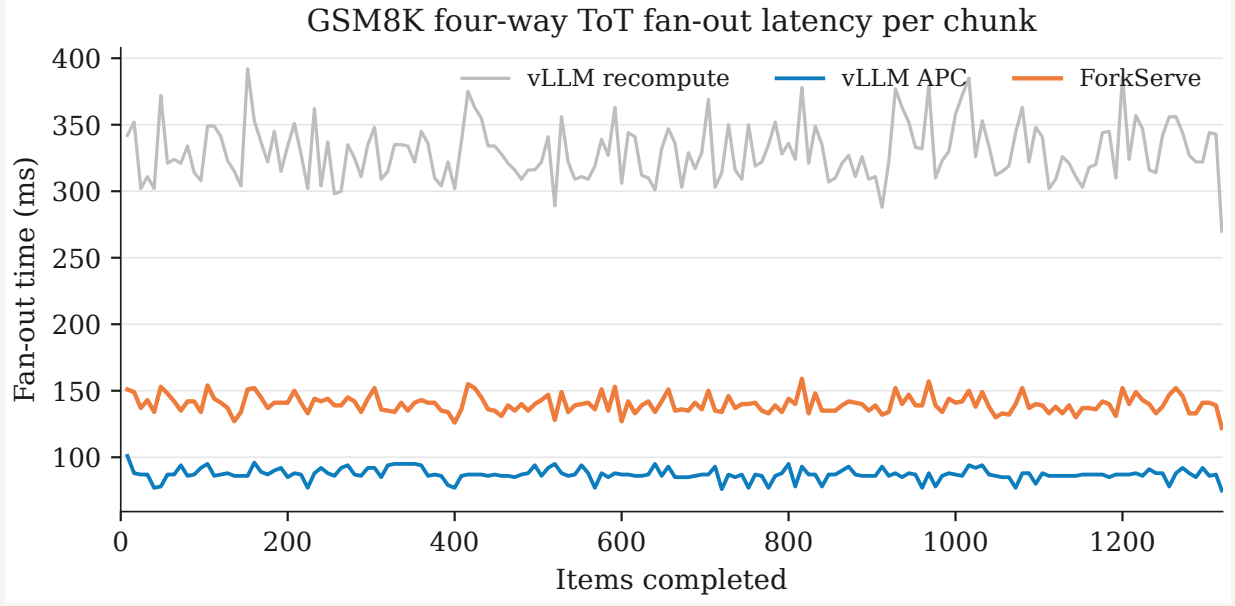


Figure 5: Per-chunk fan-out. APC is fastest; ForkServe is slower than APC but far below recompute.

5 Conclusion

On Qwen3-14B, $tp=2$, ToT forest, GSM8K test:

$$\text{peak}^{\text{FS}} = 0.72 \cdot \text{peak}^{\text{APC}} = 0.25 \cdot \text{peak}^{\text{rec}}, \quad \text{acc}^{\text{FS}} \approx \text{acc}^{\text{APC}}, \quad \frac{T_{\text{e2e}}^{\text{FS}}}{T_{\text{e2e}}^{\text{APC}}} \approx 1.016. \quad (4)$$

The $n=4$ forest reproduces the peak ordering on ToT and ReAct, including $tp=4$ Game24 and HumanEval.

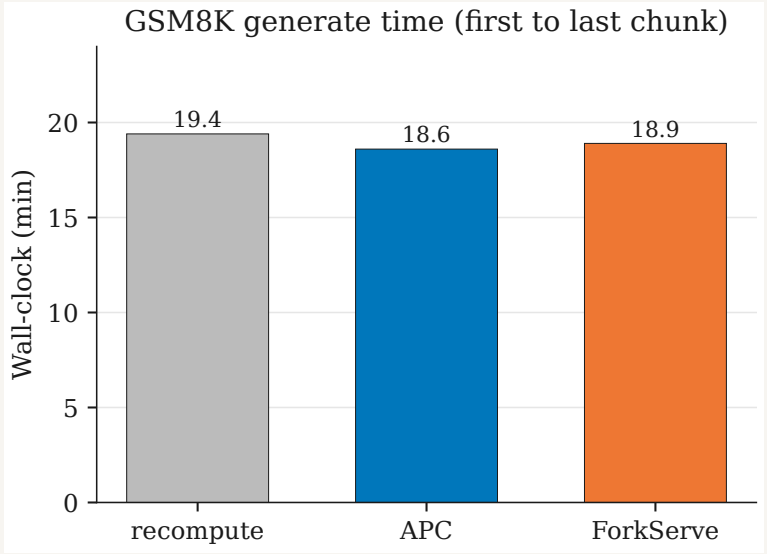


Figure 6: Generate wall-clock. ForkServe is +0.3 min vs. APC (+1.6%).

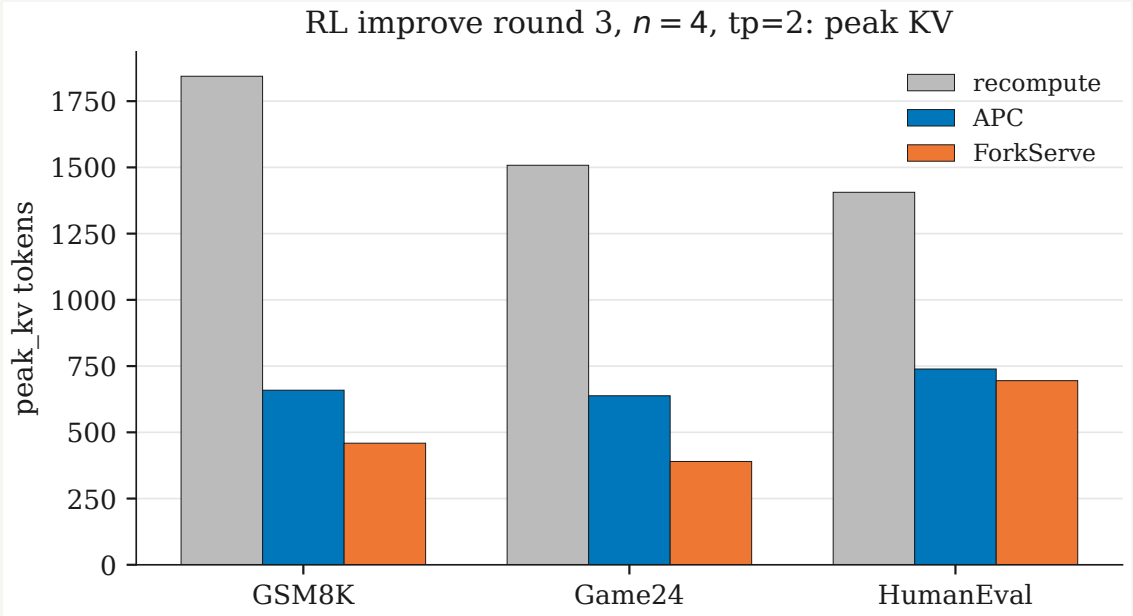


Figure 7: $tp=2$, $n=4$: peak. GSM8K 459<659, Game24 390<638, HumanEval 695<739.

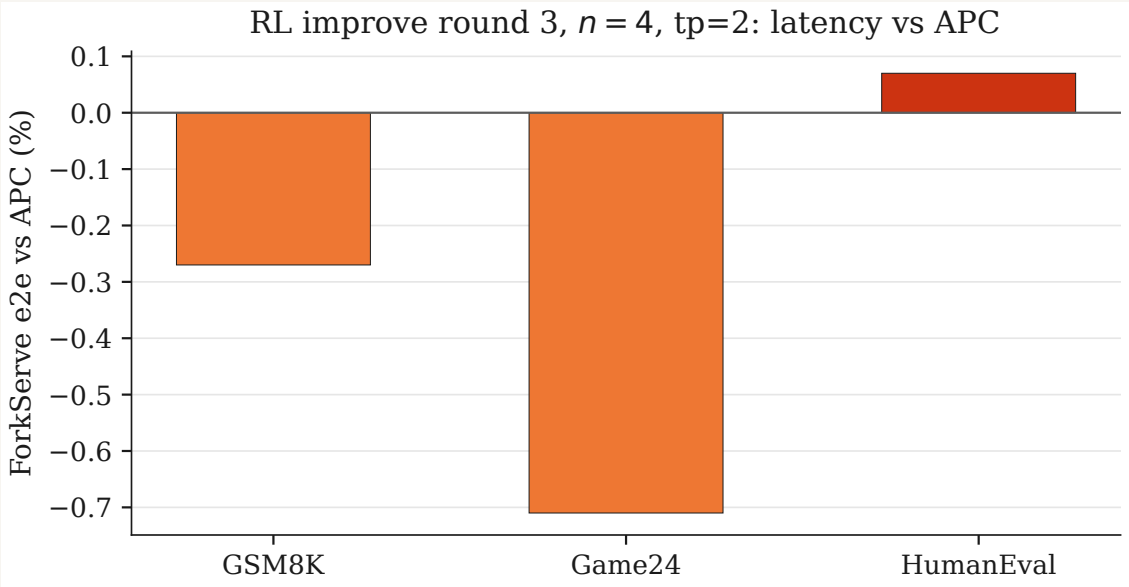


Figure 8: $tp=2$ e2e vs. APC: -0.3% , -0.7% , $+0.1\%$.

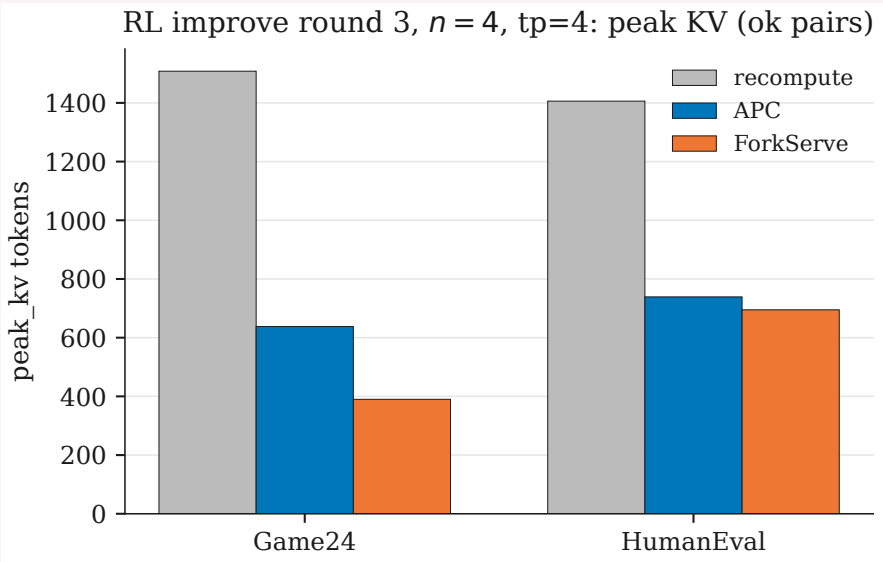


Figure 9: $tp=4$ ToT Game24 and ReAct HumanEval: same spine vs. APC gap as $tp=2$.